

1. Motivation

Traffic congestion is one of the most noticeable and important problems in Istanbul. Besides being noticeable, traffic congestion leads to wasted time, fuel consumption, economic inefficiency, and environmental pollution. In Istanbul, a megacity with a high population density, analyzing the factors affecting traffic congestion is vital for improving transportation.

While traffic congestion is often associated with inadequate infrastructure or human behavior, environmental factors, particularly weather conditions, also contribute. Drivers behavior is not the same in rainy, windy, or very hot weather.

The goal of this project is not to create a highly accurate traffic prediction model; rather, it is to examine, within the framework of learning outcomes, whether weather conditions, and especially rainfall intensity, have a significant impact on traffic congestion in Istanbul. Therefore, the project aims to draw lessons from the data and learn about the limitations of data in such complex systems.

2. Data Sources

This research uses two publicly available datasets, both obtained from the Kaggle platform.

The traffic dataset provides minute-by-minute traffic congestion indices for Istanbul. The traffic index is the primary dependent variable used in the analysis.

The weather dataset covers the years 2020–2025 and includes daily meteorological variables such as average temperature, precipitation, wind speed, and atmospheric pressure. These variables are used as features to explain traffic congestion.

This study calculates the relationship between current weather data and traffic congestion. The weather dataset used covers a long time interval and matches the traffic data appropriately in terms of dates.

3. Data Processing and Methodology

3.1 Data Preprocessing and Integration

In this project, traffic data contains minute-based timestamps, while weather data is presented daily. Therefore, ensuring temporal consistency between the two datasets is the initial step in the analysis process. For this purpose, the traffic data was reduced to a daily time scale.

First, the timestamps in the traffic dataset were converted to datetime format and defined as a time index for time series analysis. Then, the traffic data was resampled daily using the `resample('D')` function of the Pandas library. The arithmetic mean of the minute-based traffic congestion index (TI) values for each day was calculated to obtain the daily average traffic congestion. As a result of this process, high-frequency traffic observations were summarized with a single numerical traffic congestion value representing each day.

The traffic data, reduced to the daily level, was combined with daily weather data via date information, and all subsequent exploratory analyses, statistical tests, and machine learning applications were performed using this common daily dataset.

4. Exploratory Data Analysis (EDA) Results

4.1 Daily Traffic Congestion Distribution

The histogram of the daily traffic index shows a concentration between values of 28 and 38. This suggests that traffic congestion in Istanbul generally remains at moderate levels, but shows sudden upward increases on some days. These extreme values indicate that certain situations (most likely meteorological conditions or event-based factors) can lead to significant deviations from normal patterns.

4.2 Traffic Congestion by Rainfall Intensity

When traffic congestion is examined across different rainfall intensity categories, the following observations are made: median traffic congestion is lower and variability is less on days with no or light rain. In contrast, higher median traffic congestion and wider distributions are observed on days classified as moderate and heavy rainfall.

This indicates that high rainfall intensities are associated with more severe and unpredictable traffic conditions. Possible reasons for this include reduced visibility, slower driving speeds, and an increased risk of accidents.

4.3 Comparison of Asian and European Sides

When comparing the regions, it is observed that traffic congestion is higher on the European side of Istanbul compared to the Asian side. Furthermore, it has been determined that traffic congestion values on the European side are more variable and have higher extreme values. This situation may be related to the higher population density, commercial activities, and road network complexity on the European side.

4.4 Correlation

The results of the correlation analysis show a very strong positive relationship between general traffic and traffic values on both sides of the city. There is a weak and positive linear relationship between rainfall and traffic congestion. In contrast, no linear relationship is observed between traffic congestion and solar, wind speed, and atmospheric pressure.

This result shows that rainfall is positively related to traffic congestion; however, this relationship does not necessarily have to be linear, and rainfall can only affect traffic congestion beyond certain intensity levels.

5. Hypothesis Testing Results

5.1 Main Hypothesis: Precipitation

A one-way ANOVA test was applied to analyze whether traffic congestion changes between precipitation intensity levels. The test result was found to be statistically significant, and the obtained values are as follows:

$$F = 14.071$$

$$p = 5.86 \times 10^{-10}$$

Since the p-value is well below the 0.05 significance level, the null hypothesis was rejected. This result shows that the precipitation intensity variable has a significant effect on daily traffic congestion in Istanbul.

5.2 Regional Hypothesis: Asian and European Sides

ANOVA tests were applied separately for both sides of the city. The results obtained are statistically significant:

Asian Side:

$$F = 14.3505, p < 0.001$$

European Side:

$$F = 12.5162, p < 0.001$$

The results show that rainfall intensity significantly affects traffic congestion on both sides of Istanbul, but the effect may differ somewhat between regions.

5.3 Other Weather Variables

Linear regression analysis was used to evaluate the effect of other weather factors on traffic congestion. Average temperature and wind speed had a limited effect on traffic congestion; p-values of 0.0380 and 0.0384 were found for these two variables, respectively. Atmospheric pressure was found to be statistically insignificant with a value of 0.5916. These variables explain a very small portion of the variance in traffic congestion.

6. Machine Learning Results

Three regression models were used to test the predictive ability of weather variables:

Linear Regression:

$$\text{RMSE} \approx 7.08$$

$$R^2 \approx$$

Decision Tree Regressor:

RMSE $\approx$ 9.96, R^2

Random Forest Regressor:

RMSE $\approx$ 7.12, $R^2 <$

The lowest RMSE value was obtained by the Linear Regression model. However, negative R^2 values were obtained in all models; this shows that models based solely on weather variables are not sufficient to predict traffic congestion. Overfitting was observed in the Decision Tree method, the Random Forest method reduced this somewhat but did not provide a significant improvement.

7. Limitations and Future Studies

The main limitation of this study is the exclusion of many factors affecting traffic congestion that are not related to weather conditions from the model. Behavioral factors, working hours, accidents, road works, and events can play a significant role in traffic congestion; however, these were not considered in this study. The negative R^2 values obtained in the machine learning models clearly demonstrate this limitation. In future studies, the use of additional contextual variables such as holidays and accidents could improve the results of the analysis. The application of more detailed spatial traffic data and more complex models could enhance explanatory power and predictive performance.

8. References

1. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.
— Used to support the interpretation of regression models, overfitting, and predictive limitations.
2. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer.
— Referenced for bias–variance trade-off and limitations of single-feature predictive models.
3. DSA210 Lecture Slides (Weeks on EDA, Regression, and Model Evaluation).
— Concepts such as RMSE, R^2 interpretation, time-aware train–test splitting, and exploratory analysis principles are based on course slides.